Exploring the Potential of Run-Time Generative UI Through Project-Based Teaching

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Both the advantages and challenges of dynamic and adaptive interfaces have been studied for decades in HCI. On the one hand, such interfaces can accommodate diverse personal and contextual needs and help reduce cognitive load. On the other hand, they also introduce well-known challenges, including unpredictability, perceived inconsistency, and limited transparency. The emergence of run-time generative UIs appears to offer even greater power and flexibility than previous technologies. However, despite these advancements, they are likely to pursue similar goals and encounter many of the same enduring challenges.

At University of Applied Sciences Munich, we explore the potential and challenges of human-agent collaboration through a series of project-based courses in close collaboration with industry partners. Run-time generative UIs emerged as an opportunity in the course format to move beyond chat interfaces, which fail to provide the most appropriate interaction pattern for many use cases, such as quick and accurate data selection. We are currently exploring the option to let agents design UIs at runtime. In this position paper, we outline questions arising from industry conversations and students' explorations, spanning four areas: use case fit, mental models and transparency, changing practitioner roles, and implications for HCI education.

Additional Key Words and Phrases: generative UI, human-agent collaboration, research agenda

1 Introduction

Adaptive interfaces have long tried to bridge the gap between generic designs and individual user needs [1]. HCI research has explored how systems can adjust layout, content, or functionality based on user behavior, preferences, or context [4]. This work has also documented recurring challenges: adaptive systems can be unpredictable, undermining consistency; and they can be opaque, making it difficult for users to form accurate mental models.

Run-time generative UI may be an inflection point here—or simply a new method facing familiar challenges. Traditional adaptive systems select among pre-designed alternatives; generative approaches let agents compose interface elements dynamically, potentially from predefined components.

At our university, we run a course on Human-Agent Teams with six industry partners. In evaluating our last iteration, a pattern emerged: chat interfaces are ubiquitous, but fail to provide appropriate interaction patterns for many use cases. Hence, we are moving towards exploring the potential of run-time generative UI (e.g. with Google's A2UI protocol [5] and MCP Apps [6]) in the current term.

In the following, we outline questions surfaced while exploring use cases within student projects and industry conversations.

2 Research Questions

2.1 Use Case Fit

The most immediate practical question is where run-time GenUI belongs in the design space. For a given task, should we use GenUI, predefined UI, or text-only interaction?

What task characteristics predict successful use of run-time GenUI? Our early observations suggest structured input, short-lived interactions, and low brand requirements. Some tasks seem

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suited to throwaway interfaces, but users may still expect consistency across sessions—when does that expectation emerge? Which domains are categorically unsuitable—regulated industries, safety-critical contexts, brand-intensive products? Can we derive decision guidelines that help practitioners choose among these approaches?

2.2 Mental Models and Transparency

In the Intelligent User Interfaces community, mental model formation has been extensively discussed [2]. From previous work, we know both that users often have simplified or wrong mental models of AI-based systems' functioning—and that users often do not care about understanding systems as long as they work [2, 3].

Does that finding transfer to run-time GenUI, or does interface-level opacity compound agent reasoning opacity in ways that matter for trust? Do users need to know the UI is generated? Does transparency about generation affect trust or willingness to rely on the interface? These questions have analogs in earlier research, but GenUI may reframe rather than simply extend them.

2.3 Changing Practitioner Roles

Across domains, the use of generative AI changes work from producing to curating—leading to both quicker iterations and a minimum quality level, but also to "noise" and "AI slop" [8]. When interfaces are generated at design-time, how does that change the work of designers? It seems likely that they supervise or curate this process. If so, might this change a designer's identity and pride in their work? How might it impact cognitive load, ability to concentrate, and the experience of flow during the design process? Based on our observations of student projects, we hypothesize that designers need to be very deliberate with their use of GenAI to maintain quality, depth, and flow during their work.

Moving to run-time GenUI, the role of the designer shifts more fundamentally and blends into engineering domains: it might include process, systems, and component design. Additional questions in that regard include: How do teams test, evaluate, and debug dynamically generated UI [7]? Who owns accessibility and compliance for outputs no one explicitly designed?

2.4 Implications for Education

What should we teach students entering a field where GenUI may be common? If run-time generation handles layout and component selection, which design skills remain essential? Do students need "prompt engineering for UI"—and given that non-AI experts already struggle with prompt design in general [9], how do we teach this effectively? How do we teach quality judgment for generated artifacts? Our course is one experiment; we are curious what curricula others are developing.

3 What We Hope to Learn

We bring these questions to the workshop hoping to learn whether others encounter similar questions, what patterns are emerging, and where empirical research should focus first. Insights will shape what we explore next with students and industry partners.

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